

PARTHA PRATIM BHAKAT

Bhopal, India

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EDUCATION

Vellore Institute of Technology-Bhopal

2023 – 2027

B.Tech Computer Science and Engineering - **CGPA: 7.71**

Bhopal, India

PUBLICATIONS & PATENTS

- **Patent:** SELF-RECHARGING SYSTEM FOR AN UNMANNED AERIAL PLATFORM
- **Publication:** Entertainment Multimedia Compression Using YOLO V8 and GAN – *Under Review / Submitted to Springer*

PROJECTS

Multi-Language AI Translator 🗣️ | Python, Hugging Face, PyTorch, Gradio May 2026

- * **Built** an interactive multi-language translation web application utilizing **Gradio**, allowing users to seamlessly translate English text into diverse global languages.
- * Integrated Meta's **NLLB-200** (No Language Left Behind) 600M parameter model via the Hugging Face Transformers pipeline for high-accuracy neural machine translation.
- * Leveraged **PyTorch** with bfloat16 precision to optimize model inference speed and significantly reduce memory footprint during real-time translation requests.
- * **Designed** a dynamic language routing system by mapping user-friendly language inputs to strict **FLORES-200** language codes using a custom JSON registry.

AI Text-to-SQL Data Analyst | Python, Google Gemini LLM, MySQL, Agentic AI May 2026

- * **Created** an AI-powered Text-to-SQL chatbot utilizing **Google Gemini** to seamlessly translate complex natural language questions into accurate, executable MySQL statements for non-technical users.
- * Designed an end-to-end data pipeline to automate the ingestion and mapping of raw tabular data from Excel sheets into a structured **MySQL** database architecture.
- * **Constructed** a sophisticated LLM chain equipped with a custom output parser to execute generated SQL queries dynamically and format the retrieved relational data into conversational, user-friendly responses.
- * Implemented an advanced **Agentic workflow** and rigorously evaluated query generation accuracy and response fidelity utilizing the **RAGAS** (Retrieval Augmented Generation Assessment) framework.

Real-Time ASL Detection 🗣️ | Python, OpenCV, TensorFlow, Keras May 2026

- * **Developed** a real-time American Sign Language detection application capable of classifying 24 distinct alphabet gestures using a **live webcam feed**.
- * Designed and trained a custom **Convolutional Neural Network (CNN)** from scratch on the Sign Language MNIST dataset, achieving **99.8% training accuracy**.
- * Implemented batch normalization, dropout layers, and dynamic learning rate reduction algorithms to optimize model convergence and prevent overfitting.
- * Integrated **OpenCV** for real-time video processing, utilizing background subtraction and contour detection to dynamically isolate, track, and crop the user's hand for highly accurate inference.

TECHNICAL SKILLS

Languages: Python, Java, SQL, HTML5, CSS3

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face, OpenCV, LLMs, Gen AI

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, MATLAB

Database Management: MySQL, MongoDB, DBMS

Tools & Platforms: Git, GitHub, Docker, Gradio, VS Code

LEADERSHIP & EXPERIENCE

Operations Lead

Mar 2024 – Mar 2025

The Bengali Club

VIT Bhopal

- * Directed end-to-end on-ground operations and logistics, ensuring the seamless execution of club meetings and cultural gatherings.
- * Streamlined event workflows and coordinated team efforts to support the club's overarching goals.

Core Member, Marketing

May 2025 – May 2026

Anterix Club

VIT Bhopal

- * Executed targeted marketing campaigns and promotional strategies to boost club visibility and drive event registrations.
- * Collaborated with the marketing department to design and distribute engaging outreach materials across campus.

Volunteer

Aug 2024

NCC Tree Plantation Drive

VIT Bhopal

- * Contributed to a large-scale campus sustainability initiative, actively participating in the successful planting of 1,000 trees.

CERTIFICATIONS

- Complete Data Science, Machine Learning, DL, NLP Bootcamp - Udemy